

Classical Field Theory

Labix

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1 Classical Field Theory

Definition 1.0.1 (Classical Fields)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. A field is a section $s \in \Gamma^\infty(E)$.

1.1 Lagrangian Density

Definition 1.1.1 (Lagrangian Density)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. A Lagrangian density on E is a smooth bundle map

$$\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$$

Definition 1.1.2 (Lagrangian Function)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian density. Define the Lagrangian function of $\mathcal{L}$ to be the unique function $L \in \mathcal{C}^\infty(J^1(E))$ such that

$$\mathcal{L} = L\mu_\eta$$

Recall: $j^1\phi : M \rightarrow J^1E$ is the jet prolongation of ϕ .

Definition 1.1.3 (Action Functional)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $\Omega \subseteq M$ be compact. Define the action functional of $\mathcal{L}$ on Ω to be the map $S : \mathcal{C}^\infty(E) \rightarrow \mathbb{R}$ given by

$$S[\phi] = \int_\Omega \mathcal{L}(j^1\phi)$$

Definition 1.1.4 (Admissible Variations)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\Omega \subseteq M$ be compact. Define the set of admissible variations on Ω to be the set

$$\mathcal{C}_{\partial\Omega}^\infty(E|_\Omega) = \{\delta\phi \in \mathcal{C}^\infty(E|_\Omega) \mid \delta\phi|_{\partial\Omega} = 0\}$$

Definition 1.1.5 (First Variation)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $\Omega \subseteq M$ be compact. Let $\phi \in \mathcal{C}^\infty(E)$. Let $\delta\phi \in \mathcal{C}_{\partial\Omega}^\infty(E|_\Omega)$ be an admissible variation. Define the first variation of ϕ along $\delta\phi$ by

$$\delta S[\phi, \delta\phi] = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} S[\phi + \varepsilon\delta\phi] \in \mathbb{R}$$

Definition 1.1.6 (The Euler-Lagrange Operator)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Define the Euler-Lagrange operator $\text{EL} : \mathcal{C}^\infty(E) \rightarrow \mathcal{C}^\infty(\text{Hom}(E, \text{Den}(M)))$ as follows. Let $(U, x^1, \dots, x^4)$ be a chart of M . Let $(p^{-1}(U), x^i, y^j)$ be a local trivialization chart of E . Let $(p^{-1}(U), x^i, y^j, y_i^j)$ be a local chart on $J^1(E)$ where $y_i^j \in \mathcal{C}^\infty(J^1(E))$ is such that $y_i^j(j_p^1\phi) = \left. \frac{\partial y^j \circ \phi}{\partial x^i} \right|_p$. Then locally at the chart, the operator $\text{EL}(\phi)_p : E_p \rightarrow \text{Den}_p(M)$ is given by

$$\text{EL}(\phi)_p(v) = \sum_{j=1}^{\text{rank}(E)} y^j(v) \left(\frac{\partial L}{\partial y^j} \circ j^1\phi - \sum_{i=1}^4 \frac{\partial}{\partial x^i} \left(\frac{\partial L}{\partial y_i^j} \circ j^1\phi \right) \right) \Big|_p \mu_\eta$$

where $\phi \in \mathcal{C}^\infty(E)$ and $v \in E_p$.

To explain:

- For $v \in E_p$, $y^j(v)$ picks out the j th coordinate of v according to our local chart.
- $\frac{\partial L}{\partial y^j}$ is a smooth function of $J^1(E)$. We may compose with $j^1\phi : M \rightarrow J^1(E)$ to obtain a smooth function from M to $\mathbb{R}$.
- Similarly, $\frac{\partial L}{\partial y^j} \circ j^1\phi$ is a smooth function from M to $\mathbb{R}$. Using the local chart of M , we may differentiate this function with respect to x^i to obtain another smooth function from M to $\mathbb{R}$.
- Notice that everything before the Minkowski density is indeed a real number. Hence $\text{EL}(\phi)_p(v)$ is indeed an element of $\text{Den}_p(M) \cong \mathbb{R}$.

Theorem 1.1.7 (Euler-Lagrange Theorem)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $\phi \in \mathcal{C}^\infty(E)$. Then we have

$$\text{EL}(\phi) = 0$$

if and only if $\delta S[\phi, \delta\phi] = 0$ for all $\delta\phi \in \mathcal{C}_{\partial\Omega}^\infty(E|_\Omega)$.

1.2 Hamiltonian Density

Definition 1.2.1 (Conjugate Momentum Density)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $t \in \mathbb{R}$. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be an inertial foliation. Let $(p^{-1}(U), x^i, y^j, y_i^j)$ be a local chart of $J^1(E)$. Define the conjugate momentum density $\pi_t : \mathcal{C}^\infty(E) \rightarrow \mathcal{C}^\infty(\text{Hom}(E|_{\Sigma_t}, \text{Den}(\Sigma_t)))$ of $\mathcal{L}$ at t at the local chart by

$$\pi_t(\phi)_p(v) = \sum_{j=1}^{\text{rank}(E)} y^j(v) \left(\frac{\partial L}{\partial y^j} \circ j^1\phi \right) \Big|_p \mu_\eta|_{\Sigma_t}$$

Definition 1.2.2 (Hamiltonian Density)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\mathcal{L} : J^1(E) \rightarrow \text{Den}(M)$ be a Lagrangian. Let $t \in \mathbb{R}$. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be an inertial foliation. Let u_p be the corresponding inertial observer field. Define the Hamiltonian density of $\mathcal{L}$ to be the function $\mathcal{H}_t : \mathcal{C}^\infty(E|_{\Sigma_t}) \times \mathcal{C}^\infty(\text{Hom}(E|_{\Sigma_t}, \text{Den}(\Sigma_t))) \rightarrow \mathcal{C}^\infty(\text{Den}(\Sigma_t))$ given by

$$\mathcal{H}_t(\phi)_p = (\pi_t(\phi)_p(j^1\phi(p)(u_p)) - L \circ j^1\phi(p)) \mu_\eta|_{\Sigma_t}$$

Definition 1.2.3 (The Hamilton Operator)

Theorem 1.2.4 (Hamilton's Equations)

1.3 Covariant Hamiltonian Formalism

1.4 Scalar Fields and Klein-Gordon Lagrangian

Definition 1.4.1 (Klein-Gordon Lagrangian Density)

Define the Klein-Gordon Lagrangian density to be the bundle map $\mathcal{L}_{\text{KD}} : J^1(M \times \mathbb{R}) \rightarrow \text{Den}(M)$ given at local chart $(U, x^1, \dots, x^4, y, y^1, \dots, y^4)$ by

$$\mathcal{L} = \left(\frac{1}{2} \sum_{i=1}^4 \sum_{j=1}^4 \eta_{ij} y_i y_j - \frac{1}{2} m^2 y^2 \right) \mu_\eta$$

where (η_{ij}) is the matrix representation of η and $m \in \mathbb{R}$.

Lemma 1.4.2

Let $(U, x^1, \dots, x^4, y, y^1, \dots, y^4)$ be a local chart of $J^1(M \times \mathbb{R})$. Then we have

$$\mathcal{L}_{\text{KD}} = \left(\frac{1}{2} ((y_1)^2 - (y_2)^2 - (y_3)^2 - (y_4)^2) - \frac{1}{2} m^2 y^2 \right) \mu_\eta$$

Lemma 1.4.3

Let $(U, x^1, \dots, x^4, y, y^1, \dots, y^4)$ be a local chart of $J^1(M \times \mathbb{R})$. Then we have

$$\text{EL}(\phi)_p(v) = y(v) \left(-m^2 \phi(p) - \left(\frac{\partial^2 \phi}{\partial x_1^2} - \frac{\partial^2 \phi}{\partial x_2^2} - \frac{\partial^2 \phi}{\partial x_3^2} - \frac{\partial^2 \phi}{\partial x_4^2} \right) \right) \mu_\eta$$

Definition 1.4.4 (d'Alembertian)

Define the d'Alembertian operator $\square : \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M)$ by

$$\square \phi = \frac{\partial^2 \phi}{\partial x_1^2} - \frac{\partial^2 \phi}{\partial x_2^2} - \frac{\partial^2 \phi}{\partial x_3^2} - \frac{\partial^2 \phi}{\partial x_4^2}$$

Theorem 1.4.5 (Klein-Gordon Equations)

Let $\phi \in \mathcal{C}^\infty(M \times \mathbb{R})$. Then $\text{EL}(\phi) = 0$ if and only if

$$(\square + m^2)\phi = 0$$

Proposition 1.4.6 (Klein-Gordon Hamiltonian)

1.5 Noether's Theorem

Definition 1.5.1 (Continuous Symmetry)

Theorem 1.5.2 (Noether's Theorem)

2 Electromagnetism

Classical field theory is developed to accommodate discoveries in the field of electromagnetism. We demonstrate the power of this language.

The Observation of Charge

Certain objects, when rubbed together, exert forces on one another across empty space. A glass rod rubbed with silk attracts a rubber rod rubbed with fur. Two glass rods rubbed with silk repel each other. Two rubber rods repel each other.

This observation suggests that matter can acquire a property which causes it to exert and respond to forces. We call this property *electric charge*. The pattern of attraction and repulsion suggests that charge comes in two varieties. We label them positive and negative, with the convention that like charges repel and opposite charges attract.

A key further observation is that charge is *conserved*: in any isolated process, the total charge never changes. When a glass rod is rubbed with silk, charge is not created — it is transferred from one object to the other, leaving the total unchanged.

Coulomb's Law and the Electric Field

In 1785, Charles-Augustin de Coulomb measured the force between charged spheres using a torsion balance. He found that the force between two point charges q_1 and q_2 separated by a distance r is:

$$F = k_e \frac{q_1 q_2}{r^2}$$

where k_e is a constant of proportionality. The force is repulsive if the charges have the same sign, and attractive if they have opposite signs. This is Coulomb's law.

The *electric field* is the conceptual tool that makes sense of how this force is transmitted. Rather than saying that charge q_1 acts directly on charge q_2 at a distance, we say that q_1 modifies the space around it, producing an electric field $\mathbb{E}$. The charge q_2 then responds to the field it finds itself in, not directly to q_1 . The field is defined operationally as the force per unit charge exerted on a small positive test charge:

$$E(x) = \lim_{q \rightarrow 0} \frac{F(x)}{q}$$

The limit ensures we do not disturb the field we are trying to measure. The electric field is physically real: it carries energy and propagates at finite speed, so distant charges cannot influence each other instantaneously.

Magnetism and Faraday's Discovery

Magnetic phenomena were known long before electromagnetism was unified. Lodestones attract iron. Compass needles align with the Earth's field. For centuries, electricity and magnetism appeared to be entirely separate phenomena.

In 1820, Hans Christian Orsted discovered that a current-carrying wire deflects a compass needle placed nearby. A moving charge produces a *magnetic field* B in the space around it, just as a static charge produces an electric field. Andre-Marie Ampere then showed quantitatively that parallel currents attract, and anti-parallel currents repel.

In 1831, Michael Faraday made the reciprocal discovery: a changing magnetic field drives a current in a nearby wire. He described this in terms of *lines of force* threading through a loop. When the number of these lines changes — whether by moving a magnet, changing a current, or moving the loop itself — a

voltage is induced in the loop. This is electromagnetic induction.

Faraday also observed that magnetic field lines always form closed loops. No matter how a magnet is cut or arranged, one never finds an isolated magnetic “charge” (a *monopole*) from which field lines emanate. This asymmetry between electricity and magnetism — electric charges exist, magnetic charges do not — is a fundamental observational fact.

The Unification

The deepest insight of the theory, clarified by Einstein’s special relativity, is that electric and magnetic fields are not truly separate objects. They are two facets of a single *electromagnetic field*, seen from different reference frames.

An observer watching a stationary charge sees only an electric field. A second observer moving relative to the first sees the same charge in motion — a current — and therefore sees both an electric and a magnetic field. The same physical situation produces different fields depending on the observer’s state of motion. This is not a paradox; it means the split between E and B is frame-dependent, while the underlying electromagnetic field is the invariant object.

Maxwell’s Equations

Experiment 1: Coulomb’s Law: The electric field produced by a point charge q at the origin is given by the map $E : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ where

$$E(x) = \frac{q}{4\pi\epsilon_0} \frac{x}{|x|^2}$$

for some constant $\epsilon_0 \in \mathbb{R}$.

Axiom 1: Gauss’s Law: Denote ρ by the charge per unit volume of $V \subseteq \mathbb{R}^3$. From the above we can deduce

$$\int_S E \cdot dS = \frac{1}{\epsilon_0} \int_V \rho dV$$

, which equivalently says that

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}$$

Experiment 2:

$$\int_S B \cdot dS = 0$$

for a closed surface S .

Axiom 2: Gauss’s Law:

$$\nabla B = 0$$

Experiment 3: Faraday observed

$$\int_C E \cdot dl = -\frac{d}{dt} \int_\Sigma B \cdot dS$$

where C bounds the surface Σ .

Axiom 3: Faraday’s Law:

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

Experiment 4: Ampere observed???

Axiom 4: Ampere Maxwell Law:

$$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

Theorem: the above implies conservation of charge.

Lorentz force law:

$$F = q(E + v \times B)$$

Something called the field potential F packs axiom 2 and 3 into a convenient package.

Notice that all the above are not time dependent. Einstein: when talking about how these fields evolve in time, the split between E and B is frame dependent, but the electromagnetic field is frame independent. (what this means: E and B come from F , F has a frame independent description, and we recover E and B by choosing a frame)

2.1 The Potential and Deriving the Classical Maxwell Equations

Definition 2.1.1 (The Four Potential)

A four potential is a smooth section $A \in \Omega^1(M)$.

We can reverse the process: the four potential gives the sourceless maxwell equations (2) and (3). Only if we assume conservation of charge, we can imply (1) and (4).

Definition 2.1.2 (The Field Tensor)

Let $A \in \Omega^1(M)$ be a four potential. Define the field tensor of A by

$$F = dA \in \Omega^2(M)$$

Definition 2.1.3 (Electric and Magnetic Fields)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field.

- Define the electric field of A to be

$$E = \iota_u F \in \Omega^1(M)$$

Explicitly, we have $E_p(v) = F(u, v)$.

- Define the magnetic field of A to be

$$B = \iota_u * F \in \Omega^1(M)$$

Explicitly, we have $B_p(v) = (*F)(u, v)$

Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding foliation (recall that they are the slices of M along a fixed time t). Then E and B restrict to each Σ_t which are the familiar electric and magnetic fields in classical electromagnetism. The restriction does not loose information because the time component is 0.

Definition 2.1.4 (Classical Electric and Magnetic Fields)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding inertial foliation. Let $\iota_t : \Sigma_t \hookrightarrow M$ be the inclusion.

- Define the electric field at time t to be

$$E_t = \iota_t^* E \in \Omega^1(\Sigma_t)$$

- Define the magnetic field at time t to be

$$B_t = \iota_t^* B \in \Omega^1(\Sigma_t)$$

Definition 2.1.5 (The Four Current of a Potential)

Let $A \in \Omega^1(M)$ be a four potential. Define the four current to be

$$J = *^{-1} d * F \in \Omega^1(M)$$

Definition 2.1.6 (Charge Density)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Define the charge density of A by

$$\rho = \iota_u J \in \mathcal{C}^\infty(M)$$

Theorem 2.1.7 (Classical Maxwell's Equations)

Let $A \in \Omega^1(M)$ be a four potential. Let $u \in \mathcal{H}^+$ correspond to an inertial observer field. Let $\{\Sigma_t \mid t \in \mathbb{R}\}$ be the corresponding inertial foliation. Let $\iota_t : \Sigma_t \hookrightarrow M$ be the inclusion map. Then the following are true.

- Gauss's law for electricity: $d_{\Sigma_t} *_{\Sigma_t} E_t = \iota_t^* * J$.
- Gauss's law for magnetism: $d_{\Sigma_t} B_t = 0$.
- Faraday's Law: $d_{\Sigma_t} E_t + \frac{\partial B_t}{\partial t} = 0$.
- Ampere-Maxwell Law: $d_{\Sigma_t} *_{\Sigma_t} B_t - \frac{\partial E_t}{\partial t} = \iota_t^* J_{\Sigma_t}$.

2.2 The Maxwell Lagrangian

Definition 2.2.1 (The Maxwell Lagrangian)

Define the Maxwell Lagrangian to be the map $\mathcal{L}_{\text{ML}} : J^1(T^*M) \rightarrow \text{Den}(M)$ given by

$$\mathcal{L}(j^1 A)_p = -\frac{1}{4}(dA)_p \wedge *(dA)_p$$

Proposition 2.2.2

The Euler-Lagrange operator of the Maxwell Lagrangian is given by

$$\text{EL}_{\text{ML}}(A) = d * dA - *J$$

2.3 Computing the Potential From a Current

Starting with A , we defined all the other electromagnetic quantities. Physically, we start directly with the electromagnetic field and deduce all the other quantities. On the other hand, we can start with J , and ask what the other quantities are. This is more natural because physically, starting with J is the same as specifying the matter distribution (where the charges are and how they move).

Postulate: Conservation of charge: $d * J = 0$.

Definition 2.3.1 (The Four Current)

A four current is a section $J \in \Omega^1(M)$ such that $d * J = 0$.

If we started with a four potential A , recall that J is defined by the formula $d * J = d * dA$.

Proposition 2.3.2

Let $J \in \Omega^1(M)$ be a four current. Let $A \in \Omega^1(M)$. Then $d * dA = d * J$ if and only if $\text{EL}_{\text{ML}}(A) = 0$.

The problem is that we cannot just take this as the definition of the associated potential, because A is not unique. Let $f \in \mathcal{C}^\infty(M)$. Then notice that if A is a solution, then $A + df$ is also a solution. We choose a common potential as definition (so other definitions can also possibly work, but we use the following one as the standard one).

Definition 2.3.3 (Gauge Fixing)

This gives some boundary conditions so that the pde $d * J = d * dA$ is well posed.

The solution to the pde is given as follows:

Definition 2.3.4 (The Retarded Potential)

Let $J = \sum_{i=1}^4 J_i dx^i \in \Omega^1(M)$ be a four current. Define the retarded potential of J to be the one form $A = \sum_{i=1}^4 A_i dx^i$ whose components are given by

$$A_i(p_1, \dots, p_4) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{J_i(p_1 - \sqrt{(p_2 - x_2)^2 + (p_3 - x_3)^2 + (p_4 - x_4)^2}, x_2, x_3, x_4)}{\sqrt{(p_2 - x_2)^2 + (p_3 - x_3)^2 + (p_4 - x_4)^2}} dx_2 dx_3 dx_4$$

3 Relativistic Field Theory